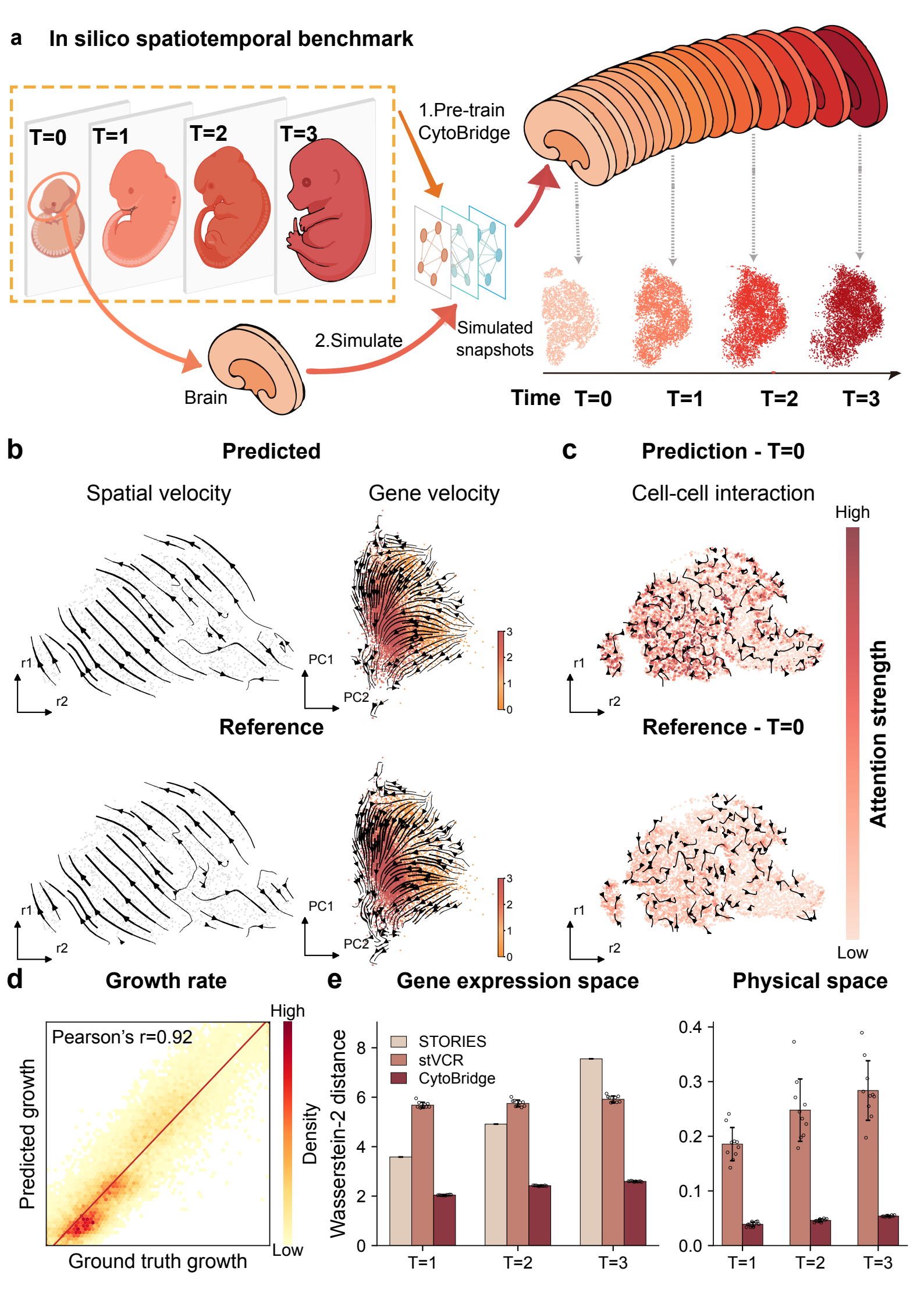

a
In silico spatiotemporal benchmark
T=0
T=1
T=2
T=3
1.Pre-train CytoBridge
2.Simulate
Brain
Simulated snapshots
Time T=0 T=1 T=2 T=3
b
Predicted
Spatial velocity
Gene velocity
r1
r2
PC1
PC2
Reference
c
Prediction - T=0
Cell-cell interaction
High
Low
Attention strength
Reference - T=0
d
Growth rate
Pearson's r=0.92
Predicted growth
Ground truth growth
High
Low
Density
e
Gene expression space
Physical space
Wasserstein-2 distance
STORIES
stVCR
CytoBridge
T=1
T=2
T=3